

Pippa Edwards

Chemical Engineer, B.E. (Hons)

Operations experienced chemical engineer (oil and gas, specialty chemicals) and doctoral researcher in heavy industry decarbonisation.

EMPLOYMENT HISTORY

Academic

February 2024 – Current (refer to ‘EDUCATION’ for details)

Research Assistant, Imperial College London, Department of Civil and Environmental Engineering (Marie-Curie Skłodowska Industry Doctoral Fellow, *DETOCS* project)

Thesis title: “Environmental and economic impact of low carbon cement”

Expected submission: November 2026

Publications:

- Edwards, P., Leal Da Silva, W.R., Fennell, P. & Myers, R.J. (2026) Strength performance and environmental assessment of Portland clinker. *Cement and Concrete Research*. 205, 108217. doi: [10.1016/j.cemconres.2026.108217](https://doi.org/10.1016/j.cemconres.2026.108217).
- Edwards, P., Dhers, S. & Myers, R.J. (2026) Life cycle assessment shows admixtures support high-performance low-carbon LC3 concrete. *Resources, Conservation and Recycling*. 235, 109095. doi: [10.1016/j.resconrec.2026.109095](https://doi.org/10.1016/j.resconrec.2026.109095).
- Edwards, P., Leal Da Silva, W.R., Fennell, P. & Myers, R.J. (2026) Toward process-based inventory data for Portland clinker substitutes. *Journal of Industrial Ecology* 1-17. doi: [10.1007/s44498-026-00186-x](https://doi.org/10.1007/s44498-026-00186-x)

Invited talks / posters:

- Speaker, "Performance-based environmental assessment of clinker quality", Gordon Research Seminar for Advanced Materials for Sustainable Infrastructure Development, Tuscany, IT, February 14 2026.
- Speaker, “Responsible material use in cement”, Responsible Raw Materials Conference, Online, April 29 2026.

Industry

February 2023 - December 2023

Operations Process Engineer, specialty chemicals & minerals processing

Key experience:

- Metallurgical gas treatment (copper sulphate smelter), sulphur solids handling and melting, sulphur furnace operations, conversion and absorption for single contact acid process, and utilities including steam boiler and water treatment systems

October 2022 – February 2023

Process Safety Engineer

Key experience:

- Delivered client report regarding loss of containment risk on offshore wellheads
- Delivered client report regarding risk and controls for handling, storing, and recharging lithium-ion cells for industrial heavy vehicles
- Competent with dispersion modelling tools (e.g., DNV PHAST and GEXCON Effects)

February 2018 – October 2022

Process Engineer, oil and gas

Roles:

- Process Engineer
- Process Control Engineer
- Process Surveillance Engineer

Key responsibilities and experience:

- Offshore processing and subsea gas wells, gas treating and liquefaction (LNG), power gen, utilities (e.g., instrument air, hot water), process and waste-water systems
- Operational risk assessments, audit, and safety related investigations
- HAZOP and LOPA
- Trip response, integrity envelope and restart monitoring, fitness for service
- Proactive surveillance of unit process performance
- Optimisation plant trials
- Management of Change (MOC)
- Process control (BPCS and APC, including monitoring, maintenance and minor improvements)
- Maintenance shutdown planning and execution support
- Production reporting
- Procedure development
- Offshore processing and subsea gas wells, gas treating and liquefaction (LNG), power gen, utilities (e.g., instrument air, hot water), process and waste-water systems

November 2016 – February 2017

Process Engineering Intern, oil and gas

November 2014 – August 2016

Process Engineering Intern, instant coffee FMCG

EDUCATION**February 2024 - Present**

Research Assistant, Imperial College London, Department of Civil and Environmental Engineering (Marie-Curie Skłodowska Doctoral Fellow, *DETOCS* project)

Thesis title: "Environmental and economic impact of low carbon cement"

Expected submission: November 2026

Key experience:

- Process-based life cycle assessment (OpenLCA, Brightway, Activity Browser, prospective LCA)
- Inventory data development (site surveys, secondary data aggregation, AspenPlus modelling)
- Thermodynamic modelling (FactSage, GEMS)
- Technoeconomic analysis (CAPEX/OPEX assessment)
- Academic consulting
- HORIZON dissemination

2014 – 2017

Bachelor of Engineering (Hons Class I) (Chemical)

University of Queensland

GPA: 6.6

Achievements / Experience

- Thesis “*Techno-economic analysis of hydrogen as a fuel source for light vehicles in Australia*”, Dow Centre for Sustainable Engineering Innovation, HD (2017)
- Tutored (CHEE2003, CHEE3003), including labs, tutorials, and marking
- Vice President, Chemical and Environmental Engineering Student Society (2017)
- Hawken Scholar (2014, 2017)
- E.S. Edminston Prize (2015)
- Dean’s Commendation for Academic Excellence (2015, 2016, 2017)
- UQ Academic Excellence Scholarship (2014 - 2017)
- Queensland Government Supporting Women Scholarship (2014 - 2017)

OTHER SKILLS

- TBOSIET & CA-EBS Certified (Nov 2021)
- Confined Space Entry (CSE) & Working at Heights accredited
- Competent with Matlab, Python, MS Office, Visio, SAP
- Competent with AspenPlus and HYSYS
- Competent with thermochemical software (FactSage, GEMS)
- Full UK working rights (UK/Australian dual citizen)

PROFESSIONAL MEMBERSHIPS AND ACTIVITIES

- RILEM Member #48276 (2025 – current)
- AIMS-Imperial Global Fellow (2024)
- Process Analytics course, Sargent Centre for Process Systems Engineering (2023)
- RedR Australia Fundamentals of Humanitarian Response course (2023)
- IChemE Affiliate Member (2018 – current)
- Engineers Australia Member (2014 – current)
- Chemical and Environmental Engineering Student Society (ChESS) Member (2014 – 2017), ChESS Volunteer Improvement Program (VIP) (2015-2017)

REFERENCES CAN BE SUPPLIED UPON REQUEST